

Finite Metric Spaces in a Lipschitz-Free Uniform Property (S) Question

Automated Math Discovery Smoke Test

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Source and Open Question

This packet concerns the Lipschitz-free item in the open-problem section of:

William B. Johnson and Tomasz Kania, *Uniform Property (S)*, arXiv:2602.09106.

The source location is page 19 of the paper.

5. OPEN PROBLEMS

Let us list open problems concerning uniform property (S).

- (1) Does c_0 have equivalent renormings with uniform property (S)?
- (2) Does every space have an equivalent renorming with uniform property (S)?
- (3) Is $U(d, a)$ always smaller than da (up to a constant)?
- (4) Characterise those pointed metric spaces whose Lipschitz-free Banach spaces have uniform property (S). For example, by Rademacher's theorem, $F([0, 1])$ is isometric to $L_1[0, 1]$ so it does have uniform property (S), unlike $F(\mathbb{N})$ that is isometric to ℓ_1 .
- (5) If X has uniform property (S), must X^{**} have it too? The converse follows from the principle of local reflexivity.
- (6) If X and Y are normed spaces and $p \in (1, \infty)$, is there a constant $C_p > 0$ so that the

The paper defines property (S) as follows: a normed space X has property (S) if for every distinct $x, y \in S_X$ and every $a > 0$ there is $z \in X$ with $\|z\| < a$ such that

$$\|x + z\| \neq \|y + z\|.$$

Uniform property (S) is the corresponding quantitative strengthening using the modulus

$$U_X(d; a) = \inf_{\substack{x, y \in S_X \\ \|x - y\| \geq d}} \sup_{\|z\| \leq a} \left| \|x + z\| - \|y + z\| \right|.$$

Uniform property (S) implies property (S).

Candidate Partial Result

Theorem 1. *Let M be a finite pointed metric space. If $|M| \geq 3$, then the Lipschitz-free Banach space $\mathcal{F}(M)$ fails property (S), and therefore fails uniform property (S).*

If $|M| = 2$, then $\mathcal{F}(M)$ has uniform property (S). If $|M| = 1$, the space $\mathcal{F}(M)$ is the zero space, so the usual unit-sphere formulation is vacuous.

Proof. We use the standard description of finite Lipschitz-free spaces:

$$B_{\mathcal{F}(M)} = \text{conv} \left\{ \frac{\delta_u - \delta_v}{d(u, v)} : u, v \in M, u \neq v \right\}.$$

Thus, when M is finite, $B_{\mathcal{F}(M)}$ is a centrally symmetric polytope. Also $\dim \mathcal{F}(M) = |M| - 1$.

Assume first that $|M| \geq 3$. Then $\dim \mathcal{F}(M) \geq 2$, so the polytope $B_{\mathcal{F}(M)}$ has a facet F of positive dimension. Choose two distinct points x, y in the relative interior of F . Let φ be a norm-one functional exposing this facet, so that

$$F = \{u \in B_{\mathcal{F}(M)} : \varphi(u) = 1\}.$$

We claim that the norm is locally equal to φ near both x and y . Indeed, because the ball is a polytope, there are finitely many norm-one functionals $\varphi_1, \dots, \varphi_N$ such that

$$\|u\| = \max_{1 \leq j \leq N} \varphi_j(u) \quad (u \in \mathcal{F}(M)),$$

and one of them is φ . Since x and y lie in the relative interior of the facet exposed by φ , all the other facet functionals are strictly smaller than φ at both x and y . By finiteness, there is $\eta > 0$ such that whenever $\|z\| < \eta$,

$$\|x + z\| = \varphi(x + z) \quad \text{and} \quad \|y + z\| = \varphi(y + z).$$

Consequently, for all $\|z\| < \eta$,

$$\|x + z\| - \|y + z\| = \varphi(x + z) - \varphi(y + z) = \varphi(x) - \varphi(y) = 0.$$

The distinct unit vectors x, y therefore violate property (S) with $a = \eta$.

If $|M| = 2$, then $\mathcal{F}(M)$ is one-dimensional. Its unit sphere consists of two opposite points, say $\{\xi, -\xi\}$. For every $0 < d < 2$ and $a > 0$, the only admissible pair is $\xi, -\xi$, and choosing $z = t\xi$ with $0 < t < \min\{a, 1\}$ gives

$$|\|\xi + z\| - \|-\xi + z\|| = 2t > 0.$$

Hence the one-dimensional space has property (S); moreover the quantitative modulus is positive for each d, a , so it has uniform property (S). $\square$

Verification Notes

No computational verifier was used for this packet. The proof is a finite-dimensional convex-geometric argument. The main review points are:

- the standard molecule description of the unit ball of $\mathcal{F}(M)$ for finite M ;
- the fact that a polyhedral norm is locally represented by a single facet functional at relative interior points of a facet;
- the one-dimensional endpoint case $|M| = 2$.

Limitations and Novelty Check

This proves only the finite pointed metric-space subcase of the Lipschitz-free open problem. It does not characterize infinite metric spaces. The result is likely folklore-level once the finite free-space unit ball is recalled, but it is a clean deterministic subcase suitable for the pipeline's first Banach-space smoke test.